MCC Entrance Test, August 2026 — Day 1 — Solutions

Level T4

Test Information. Level T4 — 180 minutes.
6 problems: 2 Easy (10 pts), 2 Medium (15 pts, two parts), 2 Hard (25 pts, three parts); 100 points total.
Areas: Algebra, Combinatorics (×2), Geometry (×2), Number Theory.
Rules: show your work (full, justified arguments earn full credit at this level); no calculators; read all six before choosing where to start; box your final answers.
A past MCC entrance test, free to try: https://nghia71.github.io/readiness

Problem 1 (10 pts).[1] Positive integers a and b satisfy $a^5 = b^4$, and positive integers c and d satisfy $c^3 = d^2$. Given that $c - a = 19$, find $d - b$.

Official solution. Since $a^5 = b^4$, every prime in a's factorization must appear to a multiple-of-4 power and every prime in b's factorization to a multiple-of-5 power, so $a = t^4$ and $b = t^5$ for some positive integer t. Likewise $c^3 = d^2$ forces $c = s^2$ and $d = s^3$ for some positive integer s.

Substituting into $c - a = 19$:

$$s^2 - t^4 = 19 \implies (s - t^2)(s + t^2) = 19.$$

Since 19 is prime and $s + t^2 > s - t^2 > 0$, the only factorization is $s - t^2 = 1$ and $s + t^2 = 19$. Adding and subtracting these gives $s = 10$ and $t^2 = 9$, so $t = 3$.

Then $a = t^4 = 81$, $b = t^5 = 243$, $c = s^2 = 100$, $d = s^3 = 1000$ (check: $c - a = 100 - 81 = 19$, as required). So

$$d - b = 1000 - 243 = \boxed{757}.$$

□

Problem 2 (10 pts).[2] Find the remainder when 2^{1990} is divided by 1990.

Official solution. Factor $1990 = 2 \cdot 5 \cdot 199$, and find 2^{1990} modulo each factor.

Modulo 2: $2^{1990} \equiv 0$.

Modulo 5: by Fermat's Little Theorem, $2^4 \equiv 1 \pmod 5$. Since $1990 = 4 \cdot 497 + 2$, $2^{1990} \equiv 2^2 = 4 \pmod 5$.

Modulo 199 **(prime)**: by Fermat's Little Theorem, $2^{198} \equiv 1 \pmod{199}$. Since $1990 = 198 \cdot 10 + 10$,

$$2^{1990} \equiv 2^{10} = 1024 \equiv 1024 - 5(199) = 29 \pmod{199}.$$

So we need x with $x \equiv 0 \pmod 2$, $x \equiv 4 \pmod 5$, $x \equiv 29 \pmod{199}$, and $0 \le x < 1990$. Writing $x = 199k + 29$ and substituting into the mod-5 condition (using $199 \equiv 4$, $29 \equiv 4 \pmod 5$):

$$4k + 4 \equiv 4 \pmod 5 \implies 4k \equiv 0 \pmod 5 \implies k \equiv 0 \pmod 5,$$

so $k = 5m$ and $x = 995m + 29$. Since 995 is odd and 29 is odd, x is even exactly when m is odd; the smallest such m is $m = 1$, giving $x = 995 + 29 = 1024$ (and $m = 3$ exceeds the range). So $2^{1990} \equiv \boxed{1024} \pmod{1990}$. □

Problem 3 (15 pts).[3] For each integer $0 \le n \le 11$, there are exactly three identical weights weighing 2^n grams. Two piles of weights are considered different if they contain different numbers of weights of some size.

(a) Suppose only the weights weighing 1, 2, and 4 grams (three of each) are available. In how many ways can a pile weighing 6 grams be formed? [4 pts]

(b) Using all twelve weights $2^0, 2^1, \ldots, 2^{11}$ (three of each), in how many ways can a pile weighing 2021 grams be formed? [11 pts]

Official solution. **Part (a).** Listing every triple (c_0, c_1, c_2) of weight-counts (sizes $1, 2, 4$) with $c_0, c_1, c_2 \in \{0, 1, 2, 3\}$ and $c_0 + 2c_1 + 4c_2 = 6$: $(0, 1, 1)$, $(0, 3, 0)$, $(2, 0, 1)$, $(2, 2, 0)$, so there are $\boxed{4}$ ways.

Part (b). Let $g(N)$ be the number of ways to form a pile weighing N grams from an unlimited supply of weights $2^0, 2^1, 2^2, \ldots$ (each usable 0, 1, 2, or 3 times); since the largest weight ever needed to reach 2021 is far below 2^{11}, the cap "$n \le 11$" never binds and can be ignored below.

In any valid pile for N, the number of weight-1 items used, c_0, must have the same parity as N (so that the rest of the pile, using only even weights, can make up the even remainder $N - c_0$).

[1] American Invitational Mathematics Examination (AIME). [2] Regional Mathematical Olympiad (RMO), India. [3] British Mathematical Olympiad, Round 1 (BMO1).

- If N is even, $c_0 \in \{0, 2\}$, leaving a remainder of N or $N-2$ to be built from weights $2, 4, 8, \ldots$; dividing that remainder by 2 reduces to the same problem, one size class down.
- If N is odd, $c_0 \in \{1, 3\}$, leaving a remainder of $N-1$ or $N-3$; dividing by 2 gives the same two reduced sizes as the even case.

In both cases, writing $h = \lfloor N/2 \rfloor$, the two remainders divided by 2 are exactly h and $h-1$. So for $N \geq 1$,

$$g(N) = g(h) + g(h-1), \qquad h = \left\lfloor \tfrac{N}{2} \right\rfloor,$$

with $g(0) = 1$ and $g(N) = 0$ for $N < 0$. Applying this repeatedly from $N = 2021$ down to the base case:

$g(0) = 1$	$g(15) = 8$	$g(126) = 64$
$g(1) = 1$	$g(29) = 15$	$g(250) = 126$
$g(2) = 2$	$g(30) = 16$	$g(251) = 126$
$g(3) = 2$	$g(31) = 16$	$g(252) = 127$
$g(5) = 3$	$g(61) = 31$	$g(503) = 252$
$g(6) = 4$	$g(62) = 32$	$g(504) = 253$
$g(7) = 4$	$g(63) = 32$	$g(505) = 253$
$g(13) = 7$	$g(124) = 63$	$g(1009) = 505$
$g(14) = 8$	$g(125) = 63$	$g(1010) = 506$

each entry following from $g(N) = g(\lfloor N/2 \rfloor) + g(\lfloor N/2 \rfloor - 1)$ applied to the two entries above it.

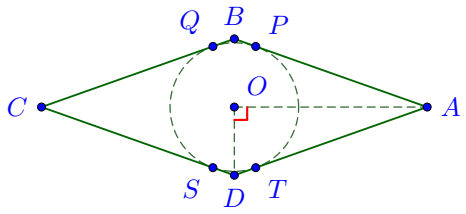
Finally, with $h = \lfloor 2021/2 \rfloor = 1010$, $g(2021) = g(1010) + g(1009) = 506 + 505 = \boxed{1011}$. □

Problem 4 (15 pts).[4]

A circle has centre O and radius 1. Quadrilateral $ABCD$ has all four sides tangent to the circle, at points P (on AB), Q (on BC), S (on CD), and T (on DA), as shown. Also, $\angle AOB = \angle BOC = \angle COD = \angle DOA$. Given that $AO = 3$,

(a) Find the length of AT. [6 pts]

(b) Find the length of DS. [9 pts]



Official solution. **Part (a).** Since DA is tangent to the circle at T, the radius OT is perpendicular to DA, so $\triangle OTA$ has a right angle at T, with $OT = 1$ (the radius) and $OA = 3$. By the Pythagorean theorem,

$$AT = \sqrt{OA^2 - OT^2} = \sqrt{9-1} = \sqrt{8} = \boxed{2\sqrt{2}}.$$

Part (b). The four angles $\angle AOB, \angle BOC, \angle COD, \angle DOA$ are equal and sum to $360°$, so each equals $90°$; in particular $\angle AOD = 90°$, so $\triangle AOD$ has a right angle at O. Since $OT \perp AD$ (part (a)), T is the foot of the altitude from the right angle O to the hypotenuse AD. By the geometric-mean (altitude-on-hypotenuse) relation,

$$OT^2 = AT \cdot TD \implies TD = \frac{OT^2}{AT} = \frac{1^2}{2\sqrt{2}} = \frac{1}{2\sqrt{2}} = \frac{\sqrt{2}}{4}.$$

Since DS and DT are both tangent segments from D to the circle, $DS = DT$, so $DS = \boxed{\dfrac{\sqrt{2}}{4}}$. □

Problem 5 (25 pts).[5] Let $ABCD$ be a convex quadrilateral with $\angle DAC = 36°$, $\angle BDC = 36°$, $\angle CBD = 18°$, and $\angle BAC = 72°$. The diagonals AC and BD intersect at point P.

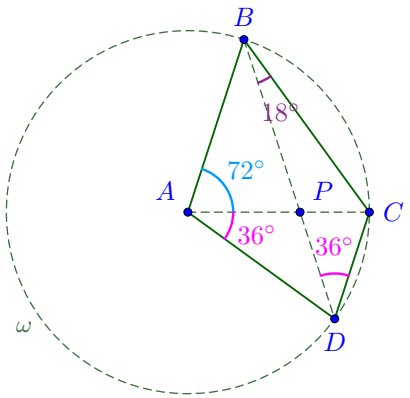
[4] Euclid Contest (CEMC, University of Waterloo).

(a) Prove that $AB = AC = AD$. [12 pts]

(b) Find $\angle ACB$ and $\angle ADC$. [3 pts]

(c) Find $\angle APD$. [10 pts]

Official solution. **Part (a).**



Let ω be the circle centered at A through C, and let B', D' be the second points where rays AB, AD meet ω.

By the Inscribed Angle Theorem applied to ω,

$$\angle B'D'C = \tfrac{1}{2}\angle B'AC = \tfrac{1}{2}\angle BAC = 36^\circ,$$
$$\angle D'B'C = \tfrac{1}{2}\angle D'AC = \tfrac{1}{2}\angle DAC = 18^\circ.$$

These match the given $\angle BDC = 36^\circ$ and $\angle CBD = 18^\circ$ exactly, and a convex quadrilateral with these four angle constraints is uniquely determined, so $B' = B$ and $D' = D$. Hence B and D both lie on ω, giving

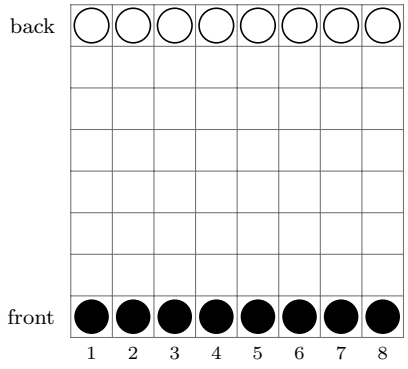
$$AB = AC = AD.$$

Part (b). By part (a), triangle ABC is isosceles with $AB = AC$ and apex angle $\angle BAC = 72^\circ$, so its base angles are $\angle ACB = \dfrac{180^\circ - 72^\circ}{2} = \boxed{54^\circ}$. Likewise triangle ACD is isosceles with $AC = AD$ and apex angle $\angle DAC = 36^\circ$, so $\angle ADC = \dfrac{180^\circ - 36^\circ}{2} = \boxed{72^\circ}$.

Part (c). In triangle APD: $\angle PAD = \angle CAD = 36^\circ$, and

$$\angle PDA = \angle BDA = \angle ADC - \angle BDC = 72^\circ - 36^\circ = 36^\circ.$$

So $\angle APD = 180^\circ - \angle PAD - \angle PDA = 180^\circ - 36^\circ - 36^\circ = \boxed{108^\circ}$. □



Problem 6 (25 pts).[6] On an 8×8 grid, eight Black tokens stand on the front rank and eight White tokens stand on the back rank. Each token moves like a queen in chess – any number of squares horizontally, vertically, or diagonally, so long as no other token blocks its path – and Black and White tokens move alternately, one token at a time.

(a) Show that at least 16 moves in total are required to interchange the two ranks. [3 pts]

(b) Show that at least 22 moves in total are required. [7 pts]

(c) Determine the minimum number of moves required with a sequence of moves achieving it. [15 pts]

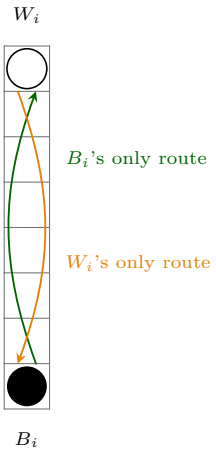
Official solution. **Part (a).** Every one of the 16 tokens starts on the rank it must vacate (Black on the front rank, White on the back rank), so each token must move at least once. Hence at least $\boxed{16}$ moves in total are required.

Part (b). Consider any of the six columns that does not contain a corner square, say column $i \in \{2, \ldots, 7\}$.

[5] Junior Balkan Mathematical Olympiad (JBMO). [6] International Mathematics Tournament of the Towns.

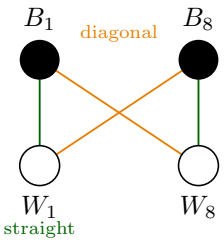
A token there can reach the opposite rank in a single move only by sliding straight along its own column: a one-move diagonal jump to the opposite rank changes the row by 7, hence also the column by 7, and this lands on the board only when the starting column is 1 or 8.

So the Black and White tokens of column i each have exactly one single-move route to the opposite rank, and it is the very square the other currently occupies – Black's only route is straight down onto White's home square $(8, i)$, White's only route is straight up onto Black's home square $(1, i)$. Whichever of the two moves first is blocked: its target is still held by its column-mate, so that move cannot land it on the opposite rank, and it needs a second move later, after its partner has vacated.



This forces at least one extra move in *each* of the six non-corner columns, so at least 6 extra moves beyond part (a)'s baseline, giving at least $16 + 6 = \boxed{22}$ moves in total.

Part (c). Now examine the four tokens standing in the two corner columns (columns 1 and 8).



Each has *two* single-move routes to the opposite rank: straight along its own column, or – since columns 1 and 8 differ by exactly 7 – diagonally to the mirror square in the other corner column. Concretely, writing squares as (row, column) with row 1 the front rank and row 8 the back rank: the Black token on $(1, 1)$ can reach $(8, 1)$ [straight] or $(8, 8)$ [diagonal]; the White token on $(8, 1)$ can reach $(1, 1)$ or $(1, 8)$; and symmetrically for the pair based in column 8. Every one of these eight candidate destinations is currently occupied by one of the *other three* corner tokens, none of which has moved yet.

So whichever corner token moves first finds *both* of its routes blocked, and it too needs a second move later – one more forced move beyond part (b)'s count: $22 + 1 = 23$.

This bound is achievable. Label each token by its starting column: B_i starts at $(1, i)$, W_i at $(8, i)$. The following 23-move sequence interchanges the ranks exactly, alternating Black and White throughout, with no two tokens ever sharing a square or crossing paths:

#	Token	Move	#	Token	Move
1	W_1	$(8,1) \to (7,2)$	13	W_5	$(8,5) \to (5,2)$
2	B_1	$(1,1) \to (8,1)$	14	B_5	$(1,5) \to (8,5)$
3	W_8	$(8,8) \to (1,1)$	15	W_6	$(8,6) \to (5,3)$
4	B_8	$(1,8) \to (8,8)$	16	B_2	$(5,6) \to (8,3)$
5	W_1	$(7,2) \to (1,8)$	17	W_7	$(8,7) \to (6,5)$
6	B_2	$(1,2) \to (5,6)$	18	B_3	$(5,7) \to (8,4)$
7	W_2	$(8,2) \to (1,2)$	19	W_7	$(6,5) \to (1,5)$
8	B_3	$(1,3) \to (5,7)$	20	B_6	$(1,6) \to (8,6)$
9	W_3	$(8,3) \to (1,3)$	21	W_5	$(5,2) \to (1,6)$
10	B_4	$(1,4) \to (3,2)$	22	B_7	$(1,7) \to (8,7)$
11	W_4	$(8,4) \to (1,4)$	23	W_6	$(5,3) \to (1,7)$
12	B_4	$(3,2) \to (8,2)$			

Moves 1–5 untangle the four corner tokens, each in turn stepping aside or completing its crossing as soon as the square ahead of it opens up. Moves 6–23 repeat the same idea for the six non-corner columns: for each pair (B_i, W_i) with $i \in \{2, \ldots, 7\}$, the currently-blocked token first sidesteps diagonally off its own column onto a temporarily empty square (moves 6, 8, 10, 13, 15, 17), which immediately frees its home square so its partner can cross directly (moves 7, 9, 11, 14, 20, 22); each side-stepped token then completes its own crossing with one further move once a target square opens up elsewhere on the rank (moves 12, 16, 18, 19, 21, 23). Checking each move in turn confirms every destination is empty when the token arrives and no path is ever blocked, and that after move 23 all eight Black tokens occupy the back rank while all eight White tokens occupy the front rank. So the minimum number of moves required is $\boxed{23}$. □